

Methods for Apportioning the Cost of a Water Resource Project

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Numerous ad hoc methods are used to apportion the costs of a water resource project among participants and/or purposes. Unfortunately these procedures do not always work, i.e., they lead to unfair assignment of costs. Accepted criteria for efficiency/financial analysis used in the water resources field are expressed in axiomatic form. These criteria are compared to accepted axioms of fairness from cooperative n -person game theory. Similarities between the two approaches are identified. The minimum costs, remaining savings (MCRS) method is proposed as an improved method of financial analysis. The MCRS method can be viewed as a generalization of the presently used separable costs, remaining benefits method.

INTRODUCTION

Major advances have been made in increasing the economic efficiency of water resource projects by taking advantage of (1) economies of scale in production and distribution facilities, (2) the assimilative capacity of the receiving environment, (3) excess capacity in existing facilities, (4) multipurpose opportunities, and (5) multigroup cooperation. The result of this economic analysis will typically indicate that the best overall solution is a relatively complex blend of management options. While this solution can be proven superior to all others from the viewpoint of economic efficiency, it may be quite difficult to implement unless an acceptable apportionment of project costs can be found. Indeed, the search for enhanced economic efficiency exacerbates the financial analysis. For example, if the problem of regionalization of sewage treatment facilities among N cities is being examined, the optimal solution may call for one large plant. In order to realize these savings, the N -communities must somehow apportion the cost of this regional facility among themselves in a 'fair' manner. As we will show, a fair solution may not exist. At the other extreme, the N cities could forego the savings due to regionalization in order to have a simple solution to the financial analysis, i.e., each city constructs and pays for its own plant. Intermediate possibilities exist. The preferred answer may be to select a 'good' solution to the economic analysis which is easy to implement from a financial point of view.

This paper reviews existing methods for examining the problem of apportioning costs among project participants in a fair manner. Then concepts from cooperative N -person game theory are offered as providing a unified conceptual framework for doing these analyses. Finally, a new method, the minimum costs, remaining savings (MCRS) method, is proposed as an improvement over existing techniques.

A note on terminology is in order before launching the discussion. Accepted practice for federal projects is to allocate costs among purposes and share costs among groups. Conceptually, these problems are the same. Thus for simplicity the problem will be referred to as one of allocating costs among groups, but 'purposes' will occasionally be used in place of 'groups.'

EVOLUTION OF CURRENT PRACTICE

The original requirements for cost allocation in the Tennessee Valley Authority project in the southeastern United

States were promulgated in 1935 [Ransmeier, 1942]. Allocation of costs was required among (1) flood control, (2) navigation, (3) fertilizer production, (4) national defense, and (5) development of power. The most controversial aspect of this problem dealt with the allocation of cost to power development. The outcome of this allocation would have a major impact on the cost of publicly produced power as compared to privately produced power, which was the primary power producer at that time.

Several prominent authorities were brought in to evaluate the numerous alternatives for allocating costs. They developed the following criteria [Ransmeier, 1942]:

1. The method should have a reasonable logical basis. It should not result in charging any objective with a greater investment than the fair capitalized value of the annual benefit of this objective to the consumer. It should not result in charging any objective with a greater investment than would suffice for its development at an alternate single purpose site. Finally, it should not charge any two or more objectives with a greater investment than would suffice for alternate dual purpose or multiple purpose improvement.
2. The method should not be unduly complex.
3. The method should be workable.
4. The method should be flexible.
5. The method should apportion to all purposes present at a multiple purpose enterprise a share in the overall economy of the operation.

The conditions described in criteria one can be represented mathematically as follows:

$$x(i) \leq \min [b(i), c(i)] \quad \forall i \in N \quad (1)$$

where

- $x(i)$ cost allocated to group i ;
- $b(i)$ benefit of group i ;
- $c(i)$ alternative cost if group i acts independently;
- N set of all groups, equal to $\{1, 2, \dots, i, \dots, n\}$.

and

$$\sum_{i \in S} x(i) \leq c(S) \quad \forall S \subset N \quad (2)$$

where $c(S)$ is the alternative cost if the subset S acts independently and S is any subset of the master set N .

Notably absent from this list of criteria is the condition that the sum of the costs allocated to the various groups equals the total cost of the project. Most of the standard

methods of cost allocation require that this condition be met. However, the TVA experts concluded that, based on economic theory, it is not possible to meaningfully allocate joint costs [Ransmeier, 1942]. The crux of their argument is that the criterion for economic efficiency requires that the marginal benefits of a purpose be compared with the marginal costs. They viewed marginal costs as being equivalent to direct costs, i.e., the costs of the elements of the project which are used solely by that purpose. Accordingly, the joint costs are viewed as sunk costs which need not be recovered. Further details about the Tennessee Valley Authority are contained in the work by *Straffin and Heaney* [1981].

With the rapid growth of federal involvement in water resources projects during the 1930's and 1940's, interagency committees were established to standardize techniques for analyzing economic and cost allocation problems. With regard to cost allocation, the committee recommended the separable costs, remaining benefits method which assigns to each group [Federal Inter-Agency River Basin Committee, 1950] (1) its separable costs and (2) a share of the remaining costs in proportion to the remaining benefits, i.e., the benefits (as limited by alternative costs) less the separable costs.

Before discussing these federal criteria, it is necessary to define 'separable costs.' They are defined as the difference between the cost of the multiple-group project and the cost of the project with the group omitted [Federal Inter-Agency, River Basin Committee, 1950]. Separable costs include direct costs and the incremental costs of changing the size of the multigroup cost elements. Thus the use of separable costs represents a significant change from the TVA procedures which used direct costs. Separable costs are

$$sc(i) = c(N) - c[(N) - \{i\}] \quad \forall i \in N \quad (3)$$

where

$sc(i)$ separable cost to group i ;
 $c(N)$ total cost for the grand coalition of n groups;
 $c[(N) - \{i\}]$ total cost for the grand coalition with group i excluded.

Given that each group has been allocated its separable costs, the remaining costs to be assigned are called nonseparable costs (nsc), or

$$nsc = c(N) - \sum_{i \in N} sc(i) \quad (4)$$

According to the remaining benefits method of apportionment each participant's prorated share of the nonseparable costs is based on the ratio of their remaining benefits (benefits of independent action less separable costs) to the total remaining benefits. This prorated share is

$$\beta(i) = [\min(b(i), c(i))] - sc(i)$$

$$\cdot \left\{ \sum_{i \in N} \{[\min(b(i), c(i))] - sc(i)\} \right\}^{-1} \quad (5)$$

where $\beta(i)$ is the prorating factor using the separable costs, remaining benefits method. Unlike the original TVA group, the *Federal Inter-Agency River Basin Committee* [1950] did require that all costs be paid. Implicit in proration formula

(5) is the requirement that

$$\sum_{i \in N} \beta(i) = 1.0 \quad (6)$$

Thus, using the separable costs, remaining benefits method, the total charge to the i th participant is

$$x(i) = sc(i) + \beta(i)nsc \quad (7)$$

James and Lee [1971] summarize other methods of apportioning project costs. With regard to a minimum charge, three options have been used: (1) set the minimum charge equal to zero, (2) set it equal to separable costs as described above, or (3) set it equal to charges unique to that activity, e.g., a canal serving an irrigation district. Having assessed this minimum charge on all participants, the remaining charges are prorated in one of six ways: (1) equally, (2) proportionally to a physical measure of use, (3) entirely to highest priority group up to limits of their benefit, (4) proportionally to benefit in excess of assigned separable cost, (5) proportionally to excess cost to provide the service by some alternate means, or (6) using the remaining benefits method described in (5). The user is left with at least these eighteen ways in which costs can be apportioned. However, no direct method has been available to permit him to rank these methods in a consistent manner. A more recent article by *Loughlin* [1977] indicates that these procedures are still being used.

COOPERATIVE N-PERSON GAME THEORY

Independent of water resources activities, game theorists have been considering problems of fairness and equity in allocating costs among members of a group who voluntarily agree to cooperate. *Rapoport* [1970] presents a good introduction to the basic ideas. *Luce and Raiffa* [1958], *Owen* [1968], and *Harsanyi* [1977] are good sources of intermediate and advanced material on this subject.

Characteristic Functions

Three types of cooperative games are presented in the literature. Games in which the players in the coalition seek to maximize profits or net revenues are called value games. Games in which the coalition of players seeks to minimize costs are called cost games. Cost games can be converted to savings games by measuring savings relative to the cost of not participating in a coalition. Thus, the savings are zero if a player chooses not to participate. These three categories of games are interrelated and lead to equivalent results. To keep the notation as simple as possible, only cost games will be discussed.

An N -person game (N, c) in characteristic function form consists of a set $N = 1, 2, \dots, n$ of players along with the characteristic function c , which assigns the real number $c(S)$ to each nonempty subset S of players. Cost games are subadditive, i.e.,

$$c(S) + c(T) \geq c(S \cup T) \quad S \cap T = \Phi \quad S, T \subset N \quad (8)$$

Φ is the empty set. S and T are any two subsets of N . If the economic optimization has been done correctly, then the subadditivity condition should be satisfied automatically, since by definition the least cost solution is used for each coalition. Satisfaction of subadditivity is a prerequisite for voluntary cooperation. If it is not satisfied, then at least one

coalition exists for which costs would be lower if the members did not form the coalition. But this is impossible if the least cost solution has been found for each coalition. At worst, no lower costs would result when the coalition formed in which case the coalition is said to be inessential, i.e.,

$$c(S) + c(T) = c(S \cup T) \quad S \cap T = \Phi \quad S, T \subset N \quad (9)$$

Fairness Criteria

Game theorists enumerate three general axioms which a fair solution to a cost game should satisfy. The first criterion states that the costs assigned to the i th group, $x(i)$, must be no more than their costs if they acted independently, i.e.,

$$x(i) \leq c(i) \quad \forall i \in N \quad (10)$$

The second criterion states that the total cost $c(N)$ must be apportioned among the n groups, i.e.,

$$\sum_{i \in N} x(i) = c(N) \quad (11)$$

All solutions satisfying criteria (10) and (11) are called imputations. The third criterion extends the first criterion by insisting that the cost to each group be no more than the costs they would receive in any coalition S contained in N , i.e.,

$$\sum_{i \in S} x(i) \leq c(S) \quad \forall S \subset N \quad (12)$$

All solutions satisfying criteria (10), (11), and (12) constitute the core of the game. For subadditive games the set of imputations is nonempty, but the core may be empty. A cost game has a convex core if

$$c(S) + c(T) \geq c(S \cup T) + c(S \cap T) \quad (13)$$

$$S \cap T \neq \Phi \quad \forall S, T \subset N$$

In general, the more attractive (lower costs) the game is, the greater the chance that the core is convex. Conversely, the less attractive the game is, the greater the chance that the core is empty.

Unique Solution Notions

The 'fairness' conditions (10), (11), and (12) significantly reduce the set of admissible solutions. However, additional criteria are needed to select a unique solution. The two most popular unique solution notions are the Shapley value [Shapley, 1953] and the nucleolus [Schmeidler, 1969]. For cost games, Shapley argues that each group should pay the incremental cost of adding them to the coalition. The incremental cost of joining a coalition is

$$c(S) - c(S - \{i\}). \quad (14)$$

Also, Shapley argues that the order in which coalitions will form is unknown. Thus he assumes that all orderings are possible and equally likely. Then the charge to the i th group, $x(i)$, is the sum of the incremental costs it pays times the probability of the various coalition formation sequences, or

$$\Phi(i) = \sum_{S \subset N} \alpha_i(S) [c(S) - c(S - \{i\})] \quad (15)$$

where

$$\alpha_i(S) = \frac{(S-1)!(n-S)!}{n!}$$

As an example, consider the following three-city cost game

$$\begin{array}{lll} c(1) = 2 & c(2) = 4 & c(3) = 6 \\ c(12) = 5 & c(13) = 6 & c(23) = 6 \\ c(123) = 7 \end{array}$$

For a three-person game there are the following six coalition formation sequences:

$$\begin{array}{lll} 123 & 213 & 312 \\ 132 & 231 & 321 \end{array}$$

Thus the Shapley value for city 1 is

$$\Phi(1) = \frac{1}{3}[c(1)] + \frac{1}{6}[c(12) - c(2)] + \frac{1}{6}[c(13) - c(3)] + \frac{1}{3}[c(123) - c(23)]$$

$$\Phi(1) = 1 \frac{1}{3}$$

Similarly, $\Phi(2) = 2 \frac{1}{3}$ and $\Phi(3) = 3 \frac{2}{3}$. Note that $\sum_i \Phi(i) = 7.0 = c(123)$.

If the game is convex, then the Shapley value is in the center of the core [Shapley, 1971]. However, for nonconvex games the Shapley value may fall outside of the core—a very undesirable property. Loehman et al. [1979] describe the Shapley value and an application to a regional wastewater management problem. They use a generalization of the Shapley value to share costs among the eight participants. Unfortunately, their resulting solution falls outside the core [Heaney, 1979].

For cost games the nucleolus maximizes the minimum savings of any coalition. The savings to a coalition, $e(S)$, are

$$e(S) = c(S) - \sum_{i \in S} x(i) \quad (16)$$

Let e_1 denote the minimum savings of any coalition S . Then

$$e_1 = \min_{S \subset N} \left[c(S) - \sum_{i \in S} x(i) \right]$$

The nucleolus can be found by solving at most $n-1$ linear programs. For the example the solution is $[1 \frac{1}{3}, 2 \frac{1}{3}, 3 \frac{2}{3}]$. The nucleolus satisfies all of the core conditions, always exists, and is unique. A disadvantage of the nucleolus is that solving $n-1$ linear programs is a bit tedious.

Comparison of Conventional Procedures to Game Theory

Striking similarities exist between the conventional procedures for allocating costs developed independently by water resources professionals and game theorists over the past fifty years [Straffin and Heaney, 1981]. Both water resources professionals and game theorists agree that participants should not be charged more than they would pay if they acted independently. While the early TVA analysts rejected the notion that full costs have to be recovered, subsequent reports supported the notion of recovery of full costs. Thus the set of imputations in game theory is in full

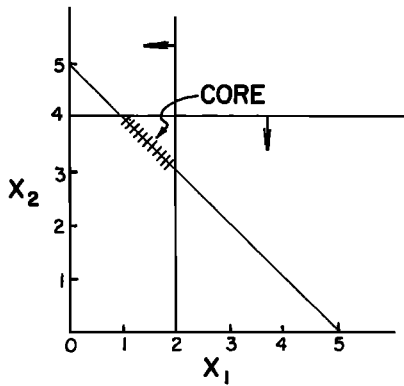


Fig. 1. Core for two-person cost game.

agreement with current practice. The very important idea of assuring that not only individuals but also any subgroup would not be left worse off was recognized by early TVA investigators and game theorists. However, this idea was not adopted in the water resources field, probably because it was felt to further complicate an already confusing situation. Thus a major contribution of the game theorists to current practice is to require that the selected answer fall in the core of the game. Detailed comparisons of conventional and game theoretic procedures are presented elsewhere [Heaney, 1979]. The remainder of this paper presents a method which is felt to be a theoretically sound and yet workable extension of existing practice in the water resources field.

Generalization of Separable Costs, Remaining Benefits (SCRB) Method

Implementation of a new idea requires careful comparison between accepted practice and the new method. The SCRB method is the most widely used approach. Thus it will be used as a basis for comparison. Three interrelated examples will be used: a two-purpose game, a three-purpose game, and a four-purpose game.

The overall idea upon which this generalization is based is to delineate the boundaries of the core. Then the minimum and maximum feasible costs for each participant are found. The minimum feasible costs correspond to the separable costs. The remaining costs are prorated based on the difference between the participant's feasible maximum and minimum costs and the total difference. The thrust of the idea will become evident as the examples are presented. The proposal method is called the minimum costs, remaining savings (MCRS) method.

Consider the following two-person cost game.

$$\begin{aligned} c(1) &= 2 & c(2) &= 4 \\ c(12) &= 5 \end{aligned}$$

The core of this game is specified by

$$\begin{aligned} x(1) &\leq 2 \\ x(2) &\leq 4 \\ x(1) + x(2) &= 5 \end{aligned}$$

A plot of this game is shown in Figure 1. Using the SCRB method, each participant pays his separable cost, $sc(1) = 1$, $sc(2) = 3$. The nonseparable cost of 1 is prorated based on

remaining savings

$$\beta(1) = \frac{2 - 1}{2} = \frac{1}{2} = \beta(2) \quad (18)$$

Using the proposed method, the core bounds on $x(1)$ and $x(2)$ are, from Figure 1, $1 \leq x(1) \leq 2$ and $3 \leq x(2) \leq 4$. But the lower bounds are simply the separable costs, and the upper bounds are identical. Indeed these answers are always identical, and the core always exists for two-person games [Heaney, 1979]. Thus cost allocation for two groups is very simple. Hence, if a reasonably good partitioning of the participants into pairs can be found, then this may be the preferred solution from an overall economic/financial analysis point of view. However, if this is not the case, then a more complicated three group situation may be evaluated. Let's add a third member to the example.

$$\begin{aligned} c(1) &= 2 & c(2) &= 4 & c(3) &= 6 \\ c(12) &= 5 & c(13) &= 6 & c(23) &= 6 \\ c(123) &= 7 \end{aligned}$$

The core conditions for this example are

$$\begin{aligned} x(1) &\leq 2 \\ x(2) &\leq 4 \\ x(3) &\leq 6 \\ x(1) + x(2) &\leq 5 \\ x(1) + x(3) &\leq 6 \\ x(2) + x(3) &\leq 6 \\ x(1) + x(2) + x(3) &= 7 \end{aligned}$$

The first three conditions set upper bounds on $x(i)$. The next three conditions set lower bounds on $x(i)$. For example, to satisfy $x(1) + x(2) \leq 5$, then $x(3) \geq 2$ in order for $\sum x(i) = 7.0$. Thus equivalent conditions are

$$\begin{aligned} 1 &\leq x(1) \leq 2 \\ 1 &\leq x(2) \leq 4 \\ 2 &\leq x(3) \leq 6 \\ x(1) + x(2) + x(3) &= 7 \end{aligned}$$

The lower bounds are simply the separable costs. Using the SCRB method, one would prorate between these upper and lower bounds and find the resulting cost allocation of $[1\frac{1}{2}, 2\frac{1}{2}, 3\frac{1}{2}]$.

Using a game theory approach, the boundaries of the core would be delimited. It is very easy to plot the three participant core using isometric paper, as shown in Figure 2. The resulting core is a quadrilateral. The objective of this analysis is to find the solution which is in the center of the core. The SCRB solution is slightly off-center. Figure 2 illustrates the real danger in using the SCRB method for three-purpose games. It prorates based on upper and lower bounds which may not be in the core. Thus, the answer not only may not be in the center of the core, it may not be in the core at all.

The proposed improvement is simply to use the upper and lower bounds as shown in the core. From Figure 2 the core

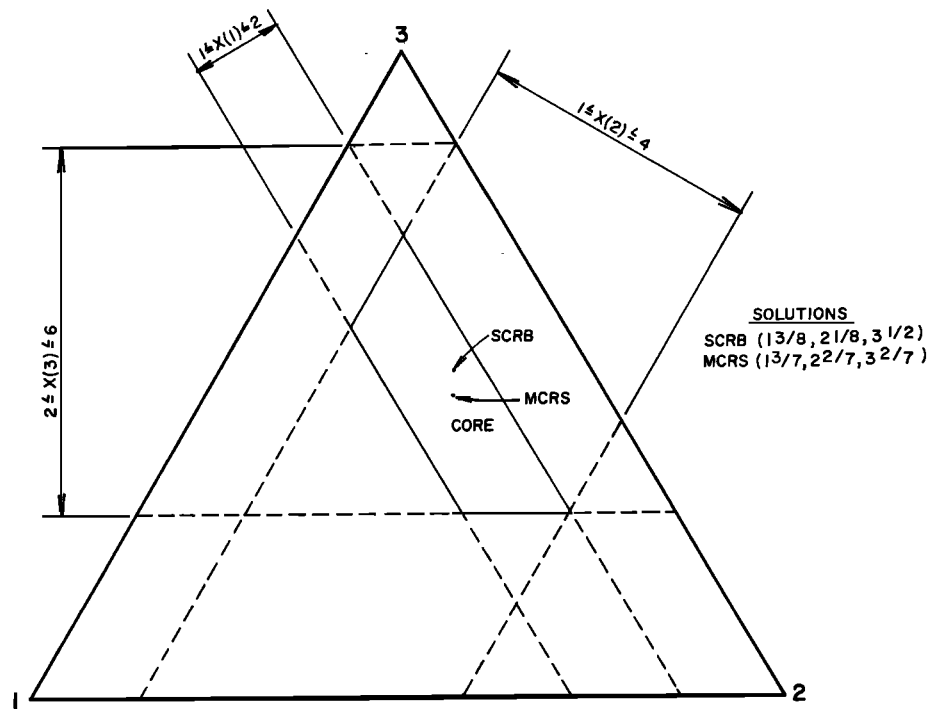


Fig. 2. Graphical solution to three-person cost game using triangular coordinate paper.

bounds are

$$1 \leq x(1) \leq 2$$

$$1 \leq x(2) \leq 4$$

$$2 \leq x(3) \leq 5$$

$$x(1) + x(2) + x(3) = 7$$

For a three-person game the core geometry can be a hexagon, pentagon, quadrilateral, triangle, line, point, or be empty. Only for the case when the game is convex (core is a hexagon) are we guaranteed that nominal bounds and actual bounds are the same. For this game the core is convex if $c(123) \leq 6.0$. On the other hand, $c(123) \leq 8.0$ to satisfy subadditivity.

For three-purpose games it is very easy to find the upper and lower bounds and the center of the core graphically. Thus the proposed extension of existing procedures for three-person cases is to find the upper and lower bounds of the core graphically and then prorate the costs as in the SCRB method. This method works fine unless the core is empty. An empty core implies that no stable solution exists. Generally, it occurs when the additional savings from forming the three member coalition are relatively small. An example of such a case is shown below.

$$\begin{array}{lll} c(1) = 4 & c(2) = 4 & c(3) = 5 \\ c(12) = 5 & c(13) = 8 & c(23) = 6 \\ c(123) = 10 \end{array}$$

These values imply the following bounds on the charges to the three groups: $4 \leq x(1) \leq 4$, $2 \leq x(2) \leq 4$, and $5 \leq x(3) \leq 5$. Given unique values of $x(1) = 4$ and $x(3) = 5$, then $x(2) = 1.0$. But this is impossible if $2 \leq x(2) \leq 4$. The SCRB method picks $[4, 1, 5]$, not a very reasonable compromise solution. A relatively straight forward way to find a compromise solu-

tion is to relax some of the bounds until a core appears. One way to do this is to relax the bounds on the intermediate S member coalitions. This problem can be formulated as a linear program, i.e.

$$\begin{array}{ll} \min & \theta \\ \text{subject to} & \\ & x(i) \leq c(i) \quad \forall i \in N \\ & \sum_{i \in S} x(i) - \theta c(S) \leq c(S) \quad \forall S \subset N \\ & \sum_{i \in N} x(i) = c(N) \\ & x(i) \geq 0 \end{array} \quad (19)$$

The optimal solution to this problem is the minimum value θ , which yields a nonempty core. Given θ , the problem can be solved graphically. For this example the optimal value of θ is 0.0526 and the revised problem is

$$\begin{array}{lll} c(1) = 4 & c(2) = 4 & c(3) = 5 \\ c(12) = 5.26 & c(13) = 8.42 & c(23) = 6.32 \\ c(123) = 10 \end{array}$$

This implies new bounds of $3.68 \leq x(1) \leq 4$, $1.58 \leq x(2) \leq 4$, and $4.74 \leq x(3) \leq 5$ and a point core of $[3.68, 1.58, 4.74]$. Additional work is needed on the problem of games without cores. However, from a practical point of view the game theory approach does alert us that a problem exists. We may decide to forego the coalition if the cost allocation problem appears to be too complicated to the decision makers in light of the relatively modest amount of economic gain to be realized. If a compromise solution is desired, then a simple linear program can be used. It is also possible to solve the

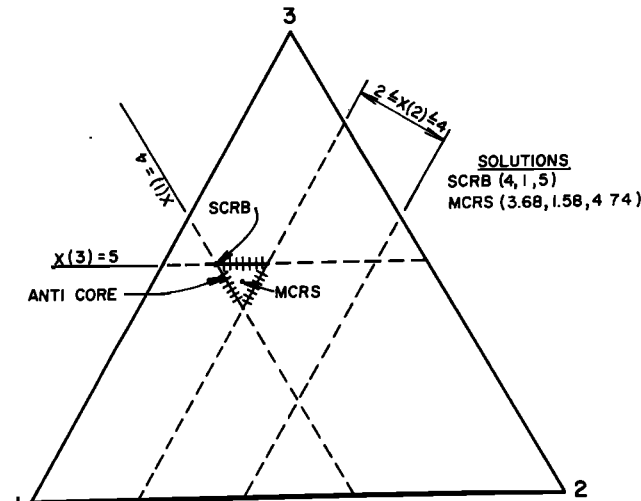


Fig. 3. Anticore of three-person game using triangular coordinate paper.

problem graphically by finding the center of the triangle forming the lower bounds (the anticore in Figure 3).

Four or More Participants

For four or more participants the financial analysis gets relatively complicated. The number of conditions to be satisfied is $2^n - 1$. Thus, even moving from a three- to four-person analysis, the number of conditions to be simultaneously satisfied goes from 7 to 15. A significant increment in effort is required to calculate costs for all subcoalitions. Furthermore, graphical solutions are no longer possible. Fortunately, linear programming methods may be used to solve even fairly complicated problems with little effort. Consider the following expanded version of the earlier two- and three-person problems:

$$c(1) = 2 \quad c(2) = 4 \quad c(3) = 6 \quad c(4) = 8$$

$$c(12) = 5 \quad c(13) = 6 \quad c(14) = 9$$

$$c(23) = 6 \quad c(24) = 10 \quad c(34) = 10$$

$$c(123) = 7 \quad c(124) = 12 \quad c(134) = 12 \quad c(234) = 11$$

$$c(1234) = 12$$

For games with a core the upper and lower bounds on each $x(i)$ can be found by solving the following linear program:

$$\max \text{ or } \min x(i)$$

subject to

$$x(i) \leq c(i) \quad \forall i \in N$$

$$\sum_{i \in S} x(i) \leq c(S) \quad \forall S \subset N \quad (20)$$

$$\sum_{i \in N} x(i) = c(N)$$

$$x(i) \geq 0 \quad \forall i \in N$$

While it is necessary to solve $2n$ linear programs, it is very easy to do since the constraint set is identical. Thus all you have to do to get additional solutions is to bring in a new

objective function and resolve the original problem. Computational details are presented by Heaney [1979].

If the game does not have a core, the solution to the linear program will be infeasible. In this case the values of the characteristic functions for the S member coalitions are relaxed until a core appears. The procedure is identical to the one outlined earlier in discussing the three-person game.

For this four-participant example the game has a core. The upper and lower bounds and the final solutions are shown in Table 1 along with the upper and lower bounds for the SCRB method. The MCRS method is preferable to the SCRB method because it uses bounds which are feasible. Thus any convex combination of these bounds is feasible. The SCRB method only analyzes coalitions of size 1, $N - 1$, and N . All other information is ignored. By contrast, the MCRS method includes all other available information regarding intermediate coalitions. Note that it is not required that all coalition values be available.

In summary, the minimum costs, remaining savings (MCRS) solution procedure is as follows:

Step 1. Find the minimum $[x(i)_{\min}]$ and maximum $[x(i)_{\max}]$ costs that satisfy the core conditions graphically or by solving linear programs 19 (no core exists) or 20 (core exists).

Step 2. Prorate the nonseparable cost as before, where

$$\beta(i) = [x(i)_{\max} - x(i)_{\min}] \left\{ \sum_{i \in N} [x(i)_{\max} - x(i)_{\min}] \right\}^{-1} \quad (21)$$

$$\forall i \in N$$

$$\text{nsc} = c(N) - \sum_{i \in N} x(i)_{\min} \quad (22)$$

Step 3. Find the fair solution for each group using

$$x(i) = x(i)_{\min} + \beta(i) (\text{nsc}) \quad (23)$$

SUMMARY AND CONCLUSIONS

The water resources literature is replete with methods for increasing economic efficiency by combining multiple purposes and/or groups in a cooperative venture. While the resultant savings are often impressive, the economic optimum may not be implemented because of difficulties in assigning the total cost among the participants in a fair manner. On the other hand, the simplest solution to implement from a cost allocation point of view is one in which all of the participants act independently. However, this can be quite inefficient. Thus the analyst needs to search not only for the optimal solution to the economic efficiency problem

TABLE 1. Comparative Results for Four-Person Cost Game

Participant	MCRS Method			SCRB Method		
	Lower Bound	Upper Bound	Solution	Lower Bound	Upper Bound	Solution
1	1	2	1.46	1	2	1.47
2	0	4	1.84	0	4	1.87
3	0	5	2.31	0	6	2.80
4	5	8	6.39	4	8	5.86
Total	6	19	12.00	5.0	20.0	12.00

but also for good solutions by which costs can be allocated in a relatively simple manner.

This paper traces the development of cost allocation guidelines in the water resources field over the past 50 years. Concurrent developments in cooperative N -person game theory are also described. These methods are combined into a more general procedure for allocating costs which is felt to be theoretically sound, computationally simple, and easy to relate to existing practice.

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REFERENCES

- Federal Inter-Agency River Basin Committee, Proposed practices for economic analysis of river basin projects, report, Washington, D.C., 1950.
- Harsanyi, J. C., *Rational Behavior and Bargaining Equilibrium in Games and Social Situations*, Cambridge University Press, New York, 1977.
- Heaney, J. P., Economic/financial analysis of urban water quality management problems, *Final Rep. EPA R802411-024*, U. S. Environ. Prot. Agency, Cincinnati, Ohio, 1979.
- Heaney, J. P., W. C. Huber, H. Sheckh, M. A. Medina, J. R. Doyle, W. A. Peltz, and J. E. Darling, Urban Stormwater Management Modeling and Decision Making, *U.S. Environ. Prot. Agency Environ. Prot. Technol. Ser., EPA-670/2-75-022*, 1975.
- James, L. D., and R. R. Lee, *Economics of Water Resources Planning*, McGraw-Hill, New York, 1971.
- Loehman, E., J. Orlando, J. Tschirhart, and A. Whinston, Cost allocation for a regional wastewater treatment system, *Water Resour. Res.*, 15(2), 193, 1979.
- Loughlin, J. C., The efficiency and equity of cost allocation methods for multipurpose water projects, *Water Resour. Res.*, 13(1), 8, 1977.
- Luce, R. D., and H. Raiffa, *Games and Decisions*, John Wiley, New York, 1957.
- Owen, G., *Game Theory*, W. B. Saunders, Philadelphia, 1968.
- Ransmeier, J. S., *The Tennessee Valley Authority*, Vanderbilt University Press, Nashville, Tennessee, 1942.
- Rapoport, A., *N-Person Game Theory*, The University of Michigan Press, Ann Arbor, Michigan, 1970.
- Schmeidler, D., The nucleolus of a characteristic function game, *SIAM J. Appl. Math.*, 17, 1163, 1969.
- Shapley, L. S., A value for n -person games, in *Contributions to the Theory of Games II, Annals Math. Studies*, vol. 28, edited by H. W. Kuhn and A. W. Tucker, Princeton University Press, Princeton, New Jersey, 1953.
- Shapley, L. S., Cores of convex games, report, p. 4620, Rand Corp., Santa Monica, Calif., 1971.
- Straffin, P. D., and J. P. Heaney, Game theory and the Tennessee Valley Authority, *Int. J. Game Theory*, 10, 1, 1981.

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